Research *Brief*

Human Authority Treaty · OCUP™ · AMD Kria KR260 Phase I

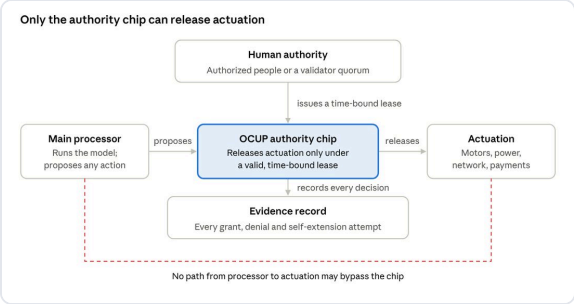


Human Authority Treaty

HAT is an international discussion proposal for coordinating expectations of machine authority while preserving national sovereignty and requiring re-evaluation when jurisdiction changes. No HAT treaty has been adopted.

OCUP architecture

One Chip Unified Protocol (OCUP) is a proposed architecture separating machine capability from permission to act, using independently governed, time-limited authority and evidence of grant, expiry, renewal, or denial. Planned features are not yet demonstrated as a completed chip.



OCUP target architecture (proposed): the intended design, not a completed chip.

Bounded Phase I result

On a physical AMD Kria KR260, a host-driven MMIO authorization sequence was granted, closed, and did not reopen during the recorded window when deliberately reasserted. STATUS 0x1060 held for five seconds, and reopen latch [4] remained clear. This was a host-driven sticky-latch control-path demonstration. It did not establish inline physical resource gating, independently sourced authority, or resistance to host, OS, DMA, or bitstream bypass.

Engineering and authorship scope

The project was organized and directed by MAXBRUCE. The published Phase I record does not establish that he individually designed or implemented the FPGA RTL. Its technical claims should be attributed to the documented experiment and its evidence, rather than to undocumented individual engineering activities.

Next research step

Evaluate non-bypassable hardware control of an accelerator or other physical resource with independent authority, measured lease expiry, and robust response to attempted bypass or replay. This is a research objective, not a completed result.